TEXORA · COURSE NOTES · LECTURE SHEET

Explaining

The skill of making things clear — the anatomy of a good explanation, the devices of defining, classifying and exemplifying, signposting language, and a worked explanation of a textile process.

HUM-1107 · COMMUNICATIVE ENGLISH · TOPIC 11 · LEVEL 1 TERM 1

1. What explaining is

To **explain** is to make something plain to someone who does not yet understand it — to carry a listener from their present knowledge to the new idea by a route they can follow. It is the teacher's, supervisor's and engineer's daily act: the trainer explaining a machine, the officer explaining a procedure to a buyer's auditor, the student explaining a process in the viva. Explaining differs from merely *telling* in one decisive respect: the explainer is responsible for the listener's understanding, not only for the accuracy of the sentences — which is why this topic closes the course's theory half and arms its practical half.

2. The anatomy of a good explanation

STAGE	WHAT THE EXPLAINER DOES
Start from the known	Begins where the listener already stands — an analogy, a familiar object, a shared experience — and builds one bridge at a time.
Give the overview first	States what the thing is and what the explanation will cover ("A sizing machine does three things…"), giving the listener a map before the journey.
Proceed in logical order	One step at a time, in the order the thing itself works — chronological for processes, part by part for objects, general to particular for ideas.
Define, classify, exemplify	Names are defined when first used; kinds are classified; every abstraction is landed on a concrete example (see §3).
Signpost the road	Uses markers — "first…", "next…", "the result is…" — so the listener always knows where they are (see §4).
Check and close	Invites questions, watches for the feedback of Topic 2, and ends with a summary that restates the whole in two or three sentences.

3. The three great devices

- **Defining:** the formal definition pattern — *term = class + distinguishing features*: “A loom (class: a machine) weaves fabric by interlacing warp and weft yarns (distinguishing features).” Loose synonyms are not definitions; this pattern is.
- **Classifying:** dividing a subject into its kinds on one stated basis at a time (fibres → natural and man-made, by origin) — the device behind most textbook chapters, including several in this very book.
- **Exemplifying and analogy:** an example gives the abstraction a face; an analogy borrows a familiar mechanism to carry an unfamiliar one (“surface tension behaves like a stretched skin” — Physics-I, Topic 3). The explainer’s honesty rule: an analogy is a ladder, not the building — state where it stops.

4. Signposting language

FUNCTION	PHRASES
Opening / overview	“Let me explain how…”, “There are three stages…”, “In simple terms…”
Sequencing	“First…”, “Then…”, “After that…”, “Finally…”
Defining	“By X we mean…”, “X is a kind of Y which…”
Exemplifying	“For example…”, “Take…”, “In other words…”
Emphasising the point	“The key point is…”, “Notice that…”
Checking	“Is that clear so far?”, “Shall I go over that part again?”
Closing	“So, to sum up…”, “In short…”

5. A worked explanation: how a dye bath exhausts

“Let me explain what happens when we dye cotton with a direct dye. There are three stages. **First**, the dye dissolves in the bath and the individual dye molecules wander freely in the water — think of them as travellers in a station. **Then**, when the cotton goes in, the molecules are attracted to the fibre surface and settle on it; this settling is called *adsorption*. **Finally**, the molecules slowly move from the surface into the fibre itself, and that inward movement — *diffusion* — is what fixes the colour where washing cannot easily remove it. The key point is this: dyeing is a journey in three steps — dissolve, settle, move in — and heat and salt are added mainly to speed the second and third steps. So, to sum up: the bath empties because the fibre collects the dye, first on its surface and then inside it.”

Mark the devices: overview with a number (“three stages”), sequence signposts, a definition at first use, an analogy (travellers in a station) doing exactly one job, a “key point” emphasis, and a closing summary that a listener could repeat back — the feedback test of Topic 2, passed.

6. Common faults

- Starting beyond the listener (jargon undefined, steps skipped) — the expert’s blind spot: knowledge makes the gaps invisible to the person who has it.
- Too much at once: five processes in one breath, with no overview and no signposts.
- No check: explaining at a nodding listener who understood nothing, because the explainer never asked.
- Analogy overload or analogy treated as proof.

7. Glossary — key terms of this topic

TERM	MEANING
Explanation	A communication that makes something understood, judged by the listener’s resulting understanding.
Definition (formal)	Statement of a term by its class and distinguishing features.
Classification	Division of a subject into kinds on a single stated basis.
Analogy	An explanation of the unfamiliar by comparison with a familiar mechanism.
Signposting	Verbal markers that show the listener the structure and progress of an explanation.

8. Key points and summary

- Explaining is measured at the listener: start from their known ground, overview first, logical order after, summary at the end, understanding checked throughout.
- Its devices: formal definition (class + features), classification on one basis, examples, and honest analogies.
- Signposting language — first/then/finally, for example, the key point is, to sum up — is the visible skeleton of every good explanation, spoken or written.

9. Practice

Answer in your own words; brief model answers follow.

1. Give formal definitions (class + distinguishing features) of: a thermometer; a shuttle; communication.
2. Classify the fibres in a typical textile syllabus on one stated basis, and name the basis.
3. Explain, in about 120 words with signposts, how a mercury thermometer measures temperature — for a school student.
4. Find the faults in this explanation to a new operator: “The machine basically does its thing through the usual mechanisms, obviously the parameters matter, it’s all standard — any questions?” (At least four.)
5. Rewrite this jargon-blocked instruction as an explanation a new helper can follow: “Ensure adequate exhaustion by optimising the liquor ratio and electrolyte dosage prior to fixation.”
6. State the six stages of §2 as a checklist and use it to plan (outline only) an explanation of “how a sewing machine forms a stitch”.
7. Why must an analogy be followed by a statement of where it stops? Answer with the “stretched skin” analogy of surface tension.

Answer key

1. Thermometer: an instrument (class) that measures temperature by the expansion of a substance, usually mercury or alcohol, in a graduated tube (features). Shuttle: a boat-shaped device (class) that carries the weft yarn through the warp shed in weaving (features). Communication: a process (class) by which a sender and receiver establish a shared meaning through messages and feedback (features).
2. Basis: origin. Fibres divide into natural fibres (from plant or animal sources — cotton, jute, wool, silk) and man-made fibres (manufactured — polyester, nylon, viscose). One basis only; mixing bases (e.g. adding “long fibres” as a third kind) breaks the classification.
3. Model qualities to show: an opening that names the thing and its parts (bulb, thin tube, mercury), a sequence (heat → mercury expands → rises in the tube → scale reading), one simple comparison allowed, and a short closing summary — all in plain words with first/then/finally signposts.
4. (1) No overview — the listener is not told what the machine does. (2) No sequence or steps. (3) Undefined jargon (“parameters”, “mechanisms”) and empty fillers (“does its thing”, “the usual”, “obviously”, “standard”). (4) No example or demonstration. (5) The closing question is rhetorical after a content-free talk — no genuine check is possible. (Any four.)
5. Model rewrite: “Before the dye is fixed, the cloth must take up as much dye as possible from the bath. To make that happen: use less water for the amount of cloth (a lower liquor ratio), and add the salt (the electrolyte) in two portions, ten minutes apart. Check the bath — when it has gone pale, most of the dye is on the cloth and fixation can begin.” — jargon defined or replaced, order given, a check supplied.

6. Checklist: start from the known; overview first; logical order; define/classify/exemplify; signpost; check and close. Outline for the stitch: known ground (needle and thread in hand sewing) → overview (two threads, one loop) → steps in order (needle descends with top thread → loop forms below → bobbin thread passes through → threads tighten) → define lockstitch at first use → signposted throughout → summary and invitation of questions.
7. Because every analogy fails somewhere, and the listener who is not told where will carry the false part with the true. The stretched-skin picture truly carries surface tension's area-shrinking and force along a line, but a real skin's tension grows when stretched while a liquid surface's tension stays constant per unit length — the point where the ladder must be left.

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